starting point. Noise describes how to construct a secure protocol from a few basic building blocks and describes a variety of handshakes that achieve different security goals. Most importantly, Noise is designed and reviewed by experts and has been used in real-world applications such as the Wireguard VPN protocol (https://www.wireguard.com).

Table 12.3 shows some recommended PSK cipher suites for TLS or DTLS 1.2 that provide forward secrecy. The ephemeral Diffie-Hellman keys can be based on either the original finite-field Diffie-Hellman, in which case the suite names contain DHE, or on elliptic curve Diffie-Hellman in which case they contain ECDHE. In general, the ECDHE variants are better suited to constrained devices because secure parameters for DHE require large key sizes of 2048 bits or more. The newer X25519 elliptic curve is very efficient and secure when implemented in software but has only recently been standardized for use in TLS 1.3.[6] The secp256r1 curve (also known as prime256v1 or P-256) is commonly implemented by low-cost secure element microchips and so is a reasonable choice too.

## Table 12.3 PSK cipher suites with forward secrecy

| Cipher suite | Description |
|---|---|
| `TLS_ECDHE_PSK_WITH_AES_128_CCM_SHA256` | PSK with ECDHE followed by AES-CCM with 128-bit keys and 128-bit authentication tags. SHA-256 is used for key derivation and handshake authentication. |
| `TLS_DHE_PSK_WITH_AES_128_CCM`<br>`TLS_DHE_PSK_WITH_AES_256_CCM` | PSK with DHE followed by AES-CCM with either 128-bit or 256-bit keys. These also use SHA-256 for key derivation and handshake authentication. |
| `TLS_DHE_PSK_WITH_CHACHA20_POLY1305_SHA256` | PSK with either DHE or ECDHE followed |